Differential Geometry

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Syllabus (One Semester)

Contents

Course Information

- **Primary texts:** Manfredo P. do Carmo, *Differential Geometry of Curves and Surfaces* (DC, 5 chapters); Barrett O'Neill, *Elementary Differential Geometry*, revised 2nd edition (ON, 8 chapters).
- **Prerequisites:** multivariable calculus, linear algebra, and basic ordinary differential equations.
- **Reading key:** [DC x-y] denotes a do Carmo section; [ON x.y] denotes an O'Neill section; bracketed author references denote supplementary texts.

1 Phase I: Curves (Fast)

Classical warm-up; build coordinate intuition quickly. Both texts cover curves in their opening chapters. This phase covers Weeks 1–2.

1.1 Week 1: Parametrized Curves and the Frenet Frame

- Parametrized regular curves, arc-length reparametrization, curvature κ and torsion τ, Frenet–Serret equations, and the fundamental theorem of curves.
- In DC, the heart is Section 1-5, "Local Theory of Curves Parametrized by Arc Length"; Sections 1-2 through 1-4 are introductory and can be read quickly.
- In ON, curves appear in Chapter 1, Section 1.4, and the full Frenet apparatus is in Chapter 2, Sections 2.2–2.3.

> **Application.** The Frenet frame is your first moving frame; the same idea underlies gauge theory via frame bundles.

Readings. [DC 1-2 through 1-5] [ON 1.4, 2.2–2.3]

Definition 1. *A Frenet frame is the moving orthonormal frame attached to a sufficiently regular space curve, consisting of tangent, normal, and binormal directions. It packages the curve's curvature and torsion into a frame-based description.*

1.2 Week 2: Global Properties of Curves (Light Treatment)

- Isoperimetric inequality sketch, Fenchel's theorem (total curvature $\geq 2\pi$), and the Four Vertex theorem (statement only).
- This material is treated lightly: global curve theory is de-emphasized in favor of surfaces.
- DC covers this in Section 1-7. ON addresses Euclidean geometry and congruence of curves in Chapter 3, Sections 3.4–3.5.

Readings. [DC 1-7 (skim)] [ON 3.4–3.5 (skim)]

> **Omitted material.** Detailed proofs of the Four Vertex theorem and global curve results; return to them later for mathematical enrichment.

2 Phase II: Surfaces in $\mathbb{R}^3$

Core of DC Chapters 2–4; ON Chapters 4–7 run in parallel. Build geometric intuition before abstraction. This phase covers Weeks 3–7.

2.1 Week 3: Regular Surfaces and Coordinate Patches

- Definition of a regular surface (DC 2-2), change of parameters (DC 2-3), tangent plane, and differential of a map (DC 2-4).
- In ON the parallel material is Chapter 4: Section 4.1 (Surfaces in $\mathbb{R}^3$), Section 4.2 (Patch Computations), and Section 4.3 (Differentiable Functions and Tangent Vectors).
- The inverse function theorem is the key technical engine throughout.

> **Application.** Coordinate patches are exactly coordinate charts on a spacetime manifold.

Readings. [DC 2-2 through 2-4] [ON 4.1–4.3]

2.2 Week 4: First Fundamental Form — The Metric Tensor

- The tensor $I = g_{ij}\, dx^i dx^j$, lengths, angles, and areas (DC 2-5).
- Orientation of surfaces (DC 2-6).
- In ON, Section 4.4 introduces differential forms on a surface; the metric and isometries are treated more fully in Chapter 6, Sections 6.4–6.5.
- Compute g_{ij} explicitly for the sphere, cylinder, and torus.

Application. The metric tensor $g_{\mu\nu}$ in general relativity is exactly this object generalized to four dimensions with indefinite signature.

Readings. [DC 2-5, 2-6] [ON 4.4, 6.4–6.5]

Definition 2. *The first fundamental form of a surface is the induced metric*

$$I = g_{ij}\, dx^i dx^j,$$

which determines intrinsic measurements such as length, angle, and area.

2.3 Week 5: Second Fundamental Form and Shape Operator

- The Gauss map and its fundamental properties (DC 3-2), which define the shape operator (Weingarten map).
- Principal curvatures κ_1, κ_2, Gaussian curvature K, mean curvature H, and the Gauss map in local coordinates (DC 3-3).
- In ON, all of Chapter 5 covers this material: Section 5.1 (Shape Operator), Section 5.2 (Normal Curvature), Section 5.3 (Gaussian Curvature), and Section 5.4 (Computational Techniques).

Readings. [DC 3-2, 3-3] [ON 5.1–5.4]

2.4 Week 6: Theorema Egregium and Gauss–Codazzi Equations

- K is intrinsic: it depends only on the first fundamental form, not the embedding.
- DC 4-3, "The Gauss Theorem and the Equations of Compatibility," contains the Gauss formula, the Theorema Egregium, and the Peterson–Mainardi–Codazzi equations.
- In ON, Chapter 6, Sections 6.1–6.2 cover the same ground using differential forms.
- Work through both proofs; they illuminate each other.

Application. Gauss–Codazzi equations are the prototype for Bianchi identities in general relativity.

Readings. [DC 4-3] [ON 6.1–6.2]

Theorem 1 (Theorema Egregium)**.** *The Gaussian curvature of a regular surface is an intrinsic invariant: it is determined by the first fundamental form alone.*

2.5 Week 7: Parallel Transport, Geodesics, and the Covariant Derivative

- DC 4-4, "Parallel Transport; Geodesics," covers covariant differentiation of a vector field along a curve, parallel transport, geodesic equations, and Christoffel symbols Γ^k_{ij}.
- Solve geodesic equations explicitly for the sphere and cylinder.
- In ON, Section 2.5 introduces covariant derivatives in $\mathbb{R}^3$; the intrinsic version on surfaces is in Sections 7.3–7.4.

Application. The geodesic equation is the equation of motion in general relativity. Christoffel symbols are the gravitational connection coefficients.

Readings. [DC 4-4] [ON 2.5, 7.3–7.4]

Omitted material. DC 4-5 (Gauss–Bonnet) and ON 7.6–7.7. Omitted to save time; Goal is to understand the theorem, and need to do a deep-dive again closer to beginning topology.

3 Phase III: Abstract Manifolds and Curvature Tensor

The language of modern theoretical physics. ON Chapter 4 bridges surfaces to manifolds; Lee supplements for the full abstract treatment. DC's five-chapter book does not cover abstract manifolds or the Riemann curvature tensor; those topics are in do Carmo's separate *Riemannian Geometry* text. DC 5-10 gives a brief bridge through abstract surfaces and further generalizations. This phase covers Weeks 8–11.

3.1 Week 8: Smooth Manifolds, Tangent Spaces, and Vector Fields

- ON 4.8 introduces the abstract manifold definition in the context of surfaces, which gives a natural bridge from the embedded picture.
- For a full treatment, use John M. Lee, *Introduction to Smooth Manifolds* (ISM), Chapters 1–3 on manifolds, smooth maps, and tangent spaces.
- DC 5-10 gives a brief, readable sketch of abstract surfaces as motivation.

Application. Spacetime is a four-manifold. Every classical and quantum field theory lives on one.

Readings. [ON 4.7–4.8] [DC 5-10 (sketch)] [Lee ISM Chs. 1–3 (primary)]

3.2 Week 9: Tensors, Differential Forms, and the Exterior Derivative

- ON Chapter 1, Sections 1.5–1.6 introduce one-forms and differential forms on $\mathbb{R}^3$; ON 4.4 extends this to surfaces.
- For the full tensor calculus—covectors, (p, q)-tensors, wedge product, exterior derivative d, and pullbacks—use Theodore Frankel, *The Geometry of Physics*, Chapter 2 as the primary physics-oriented reference.
- Practice index notation and coordinate-free language side by side.

Application. Maxwell's equations are $dF = 0$ and $d \star F = J$. Yang–Mills theory lives in this language.

Readings. [ON 1.5–1.6, 4.4] [Frankel Ch. 2 (primary for tensors)]

3.3 Week 10: Connections and the Levi-Civita Connection

- ON 2.5 (Covariant Derivatives in $\mathbb{R}^3$) and 2.7–2.8 (Connection Forms, Structural Equations) provide the Cartan-form approach to connections.
- ON 6.1 ties this to surface geometry.
- For the abstract Levi-Civita connection on a Riemannian manifold—torsion-free, metric-compatible, uniqueness theorem, and Christoffel symbols from g—use Lee, *Riemannian Manifolds* (RM), Chapters 4–5.

Application. Gravity in general relativity is entirely encoded in the Levi-Civita connection of the spacetime metric.

Readings. [ON 2.5, 2.7–2.8, 6.1] [Lee RM Chs. 4–5 (Levi-Civita)]

3.4 Week 11: Riemann Curvature Tensor, Ricci Tensor, and Bianchi Identities

- ON 7.2 (Gaussian Curvature on a geometric surface) and 7.3 (Covariant Derivative) connect K to the abstract curvature tensor.
- For the full Riemann tensor R as a $(1, 3)$-tensor, sectional curvature $K(\sigma)$, Ricci tensor $\mathrm{Ric} = \mathrm{tr}\, R$, scalar curvature R, and the first and second Bianchi identities, use Lee RM Chapter 7 or Robert Wald, *General Relativity*, Chapter 3.
- Reconcile this with the Gaussian curvature K computed in Phase II.

Application. Einstein's field equations are $\mathrm{Ric} - \frac{1}{2}Rg = 8\pi GT$. At this point, you have all the mathematical pieces.

Readings. [ON 7.2–7.3] [Lee RM Ch. 7 or Wald GR Ch. 3 (primary)]

4 Phase IV: Lorentzian Geometry and GR Bridge

O'Neill's *Elementary Differential Geometry* does not cover Lorentzian or semi-Riemannian geometry; that is the subject of his separate book *Semi-Riemannian Geometry with Applications to Relativity* (SRG). Primary texts for this phase are O'Neill SRG and Wald GR. This phase covers Weeks 12–14.

4.1 Week 12: Indefinite Metrics and Minkowski Space

- Semi-Riemannian metrics of signature (p, q), spacelike/timelike/null vectors, raising and lowering indices, and the Minkowski metric $\eta_{\mu\nu}$.
- Causal structure: light cones, causal future, and causal past.
- O'Neill SRG Chapters 1–2 give the clean abstract treatment. Wald GR Chapter 2 is a concise physics-side entry point.

> **Application.** Minkowski space is the local model for every event in a Lorentzian spacetime. Special relativity lives here.

Readings. [O'Neill SRG Chs. 1–2] [Wald GR Ch. 2]

4.2 Week 13: Geodesics in Lorentzian Manifolds and Schwarzschild Spacetime

- Timelike, spacelike, and null geodesics.
- Geodesic deviation and the Jacobi equation.
- The Schwarzschild metric: derive its geodesic equations, compute light deflection and perihelion precession as applications.
- O'Neill SRG Chapters 3 and 5 cover semi-Riemannian geometry and geodesics. Wald GR Chapters 3 and 6 cover curvature and the Schwarzschild solution.

> **Application.** Light rays are null geodesics; freely falling particles are timelike geodesics. This is the geometry of gravity.

Readings. [O'Neill SRG Chs. 3, 5] [Wald GR Chs. 3, 6]

4.3 Week 14: Synthesis and Outlook

- Review the full logical chain:

$$g \longrightarrow \Gamma \longrightarrow R \longrightarrow \mathrm{Ric} \longrightarrow \text{Einstein field equations.}$$

- Discuss which results from Phases I–II—Gauss–Bonnet, Theorema Egregium, and geodesic equations on surfaces—reappear in the four-dimensional Lorentzian setting.
- Close with a brief orientation to next steps.

Readings. [O'Neill SRG (review)] [Next: Nakahara or Wald GR Ch. 4+]

5 Final Assessment

- A final expository project connecting differential geometry with undergraduate quantum mechanics.
- Primary goal: begin with the standard quantum-mechanical formulation of the Aharonov–Bohm effect and systematically reformulate the system using the language of differential forms, connections, curvature, gauge fields, and holonomy.
- The project should culminate in a concise expository paper emphasizing the geometric meaning of gauge potentials, parallel transport, and topological phase shifts.
- Expected components may include:
  - Review of the Schrödinger equation and electromagnetic potentials.
  - Differential forms and the electromagnetic 2-form.
  - Connections and covariant derivatives.
  - Holonomy and geometric phase.
  - Interpretation of the Aharonov–Bohm phase as a holonomy phenomenon.
- Suggested outcome: a polished 6–10 page exposition synthesizing undergraduate quantum mechanics with the geometric language developed throughout the course.

6 Topics Covered

- **Curves:** Parametrized curves in $\mathbb{R}^3$, arc length, curvature and torsion, Frenet–Serret frame, fundamental theorem of curves, and global properties.
- **Surfaces in $\mathbb{R}^3$:** Regular surfaces, coordinate patches, tangent plane, first fundamental form (metric tensor), second fundamental form, Gauss map, shape operator, principal/Gaussian/mean curvatures, Theorema Egregium, Gauss–Codazzi compatibility equations, parallel transport, Christoffel symbols, and geodesics.
- **Abstract manifolds and curvature:** Smooth manifolds and atlases, tangent spaces, vector fields, Lie bracket, differential forms and exterior derivative, covariant derivatives, connections, Levi-Civita connection, Riemann curvature tensor, Ricci tensor, scalar curvature, and Bianchi identities.
- **Lorentzian geometry:** Semi-Riemannian metrics, Minkowski space, causal structure, null/timelike/spacelike geodesics, Jacobi fields and geodesic deviation, Schwarzschild metric and its geodesics, and Einstein field equations.

7 Recommended Supplements

- Frankel, *The Geometry of Physics*: physics notation alongside mathematics; ideal for Weeks 9–11.
- Lee, *Introduction to Smooth Manifolds*: rigorous abstract manifold foundations; primary for Week 8.
- Lee, *Introduction to Riemannian Manifolds*: connections and curvature tensors in full generality; primary for Weeks 10–11.
- O'Neill, *Semi-Riemannian Geometry with Applications to Relativity*: primary for Phase IV.
- Wald, *General Relativity*: natural next text after this course; used in Weeks 11–14.
- Nakahara, *Geometry, Topology and Physics*: fiber bundles, gauge theory, and instantons; recommended next step after this course.